Qs 1

Solution

State Representation:

- (4-gallon jug, 3-gallon jug)

Step	Action	State
1	Initial State	(0,0)
2	Fill 3-gallon jug	(0,3)
3	Pour into 4-gallon jug	(3,0)
4	Fill 3-gallon jug again	(3,3)
5	Pour into 4-gallon jug	(4,2)
6	Empty 4-gallon jug	(0,2)
7	Pour remaining water into 4-gallon jug	(2,0)

Goal achieved.

Qs 2

Solution

(i) Percepts

The rover receives information from:

- Cameras
- Temperature sensors
- Rock and soil sensors
- Distance sensors
- GPS/Navigation sensors
- Battery level sensor

(ii) Operating Environment

- Partially Observable
- Dynamic
- Sequential
- Continuous
- Unknown
- Real-world environment

(iii) Actions

- Move Forward
- Move Backward
- Turn Left
- Turn Right
- Capture Images
- Collect Soil Sample
- Analyze Rock
- Send Data to Earth
- Avoid Obstacles
- Recharge (Power Management)

(iv) Performance Measure

The rover performs well if it

- Collects maximum samples
- Avoids obstacles
- Uses minimum energy
- Reaches target safely
- Sends accurate information

(v) Suitable Agent Architecture

Goal-Based Agent

Reason:

- It has a specific goal (explore Mars).
- It plans actions to achieve the goal.
- It adapts to unknown environments.

Qs3

Solution

Initial State

The chessboard is empty.

State Space

All possible arrangements of queens on the chessboard.

Operators

- Place one queen in a safe position.
- Move to the next row.
- Backtrack if a conflict occurs.

Goal State

Eight queens are placed without attacking each other.

Search Strategy

Backtracking Search (Depth First Search)

Reason:

- Place queens one by one.
- If a queen attacks another queen, remove it (Backtrack).
- Continue until all 8 queens are safely placed.

Qs 4

Solution

Problem-Solving Agent

Goal-Based Agent

Reason

The customer's goal is to:

- Reach the destination safely.
- Select the best cab.
- Minimize travel cost and time.

The agent compares different cab options and chooses the most suitable one.

Pseudocode

Start

Input Pickup Location

Input Destination

Input Cab Type

IF Cab Available THEN

    Calculate Fare

    Book Cab

    Display Driver Details

ELSE

    Display "Cab Not Available"

END IF

Stop

Qs 5

Solution

Uniform Cost Search (UCS) is an uninformed search algorithm that always expands the node with the **lowest cumulative path cost**.

Step-by-Step Execution

Step 1: Start at **S** (Cost = 0)

Possible paths:

- S → A = 1
- S → G = 12

Choose **A** (lowest cost).

Step 2: Expand **A** (Cost = 1)

Possible paths:

- S → A → B = 1 + 3 = 4
- S → A → C = 1 + 1 = 2

Choose **C** (lowest cost).

Step 3: Expand **C** (Cost = 2)

Possible paths:

- S → A → C → D = 2 + 1 = 3
- S → A → C → G = 2 + 2 = 4

Choose **D** (lowest cost).

Step 4: Expand **D** (Cost = 3)

Possible path:

- S → A → C → D → G = 3 + 3 = 6

The priority queue now contains:

- S → A → B = 4
- S → A → C → G = 4
- S → A → C → D → G = 6
- S → G = 12

The next lowest-cost goal node is **S → A → C → G**.

Cost Calculation

Path	Cost
S → G	12
S → A → B → D → G	10
S → A → C → D → G	6
S → A → C → G	**4**

Optimal Path: S → A → C → G

Minimum Cost: 4